

MuddyWater Threat Intelligence Investigation Report

Threat Intelligence Analysis and Threat Actor Profiling Using MISP

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Organization: SecureOps Cybersecurity Inc.

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Threat Actor Investigated

MuddyWater (APT)

Also known as:

- Mango Sandstorm (Microsoft)
- MERCURY
- Seedworm
- Static Kitten
- TEMP.Zagros

Project Focus

- Cyber Threat Intelligence
- Security Operations Center (SOC)
- Threat Hunting

Project Type

- Threat Intelligence Investigation
- Threat Actor Profiling
- IOC Analysis
- MITRE ATT&CK Mapping

Technologies Used

- MISIP (Malware Information Sharing Platform)
- Docker
- Kali Linux
- MITRE ATT&CK Framework

2. Executive Summary

SecureOps Cybersecurity Inc. is a global cybersecurity organization headquartered in Europe, specializing in threat intelligence, incident response, risk analysis, and vulnerability management for government agencies, financial institutions, healthcare providers, energy companies, and other critical infrastructure organizations. As the threat landscape continues to evolve, SecureOps has observed a significant increase in sophisticated cyberattacks attributed to nation-state Advanced Persistent Threat (APT) groups targeting high-value organizations across Europe. These attacks have demonstrated increasing levels of sophistication through targeted phishing campaigns, malware deployment, credential theft, and supply chain compromise, creating significant challenges for timely threat detection and accurate threat attribution.

To strengthen its threat intelligence capabilities and improve proactive defense strategies, SecureOps initiated this project to investigate **Team 5 – MuddyWater**, a well-known nation-state threat actor associated with cyber espionage operations. The investigation leveraged the **Malware Information Sharing Platform (MISP)** to collect, correlate, and analyze real-world threat intelligence obtained from multiple community threat feeds. By consolidating threat intelligence within MISP, the project aimed to identify Indicators of Compromise (IOCs), profile the threat actor, analyze attack infrastructure, and provide actionable intelligence that would enable client organizations to improve their overall security posture.

The project involved deploying MISP within a Docker containerized environment on Kali Linux, configuring default threat intelligence feeds, synchronizing threat data, and conducting a comprehensive investigation of MuddyWater-related events. The investigation focused on identifying malicious infrastructure, extracting indicators such as IP addresses, domains, URLs, registry artifacts, and cryptographic file hashes, and correlating these indicators with existing intelligence available within MISP. Additional analysis was performed to examine the threat actor's operational characteristics, known aliases, geographic attribution, motivations, common tactics, techniques, and procedures (TTPs), notable campaigns, and associated malware.

Threat intelligence obtained during the investigation identified **MuddyWater**, also publicly tracked as **Mango Sandstorm**, **MERCURY**, **Seedworm**, **Static Kitten**, and **TEMP.Zagros**, as an Iranian state-aligned Advanced Persistent Threat (APT) group primarily engaged in long-term cyber espionage campaigns. The investigation revealed the group's continued reliance on spear-phishing techniques, living-off-the-land binaries (LOLBins), PowerShell-based execution, credential theft, command-and-control infrastructure, and persistence mechanisms designed to maintain long-term access to compromised environments. These observed behaviors aligned with multiple techniques documented within the **MITRE ATT&CK®** framework, providing valuable insight into the adversary's operational methodology.

The analysis further demonstrated the effectiveness of MISP as a collaborative threat intelligence platform capable of correlating threat events, enriching Indicators of Compromise, and supporting threat attribution through integrated threat intelligence feeds and Galaxy clusters.

By examining correlated events and threat actor relationships, the project successfully established connections between MuddyWater campaigns and associated threat infrastructure, enabling a deeper understanding of the adversary's operational patterns.

The findings of this investigation provide SecureOps Cybersecurity Inc. with actionable threat intelligence that can be incorporated into Security Operations Center (SOC) workflows, threat hunting activities, incident response procedures, and detection engineering. The extracted Indicators of Compromise can be operationalized through SIEM platforms, endpoint detection solutions, and network security controls to improve early detection and response to MuddyWater-related activities. Furthermore, the developed threat actor profile serves as a reusable intelligence resource that supports future investigations involving similar nation-state threat groups.

Overall, this project demonstrates how structured threat intelligence collection, correlation, and analysis using MISP can significantly improve an organization's ability to identify sophisticated adversaries, attribute malicious activity, and implement proactive defensive measures against emerging cyber threats targeting critical infrastructure and government organizations.

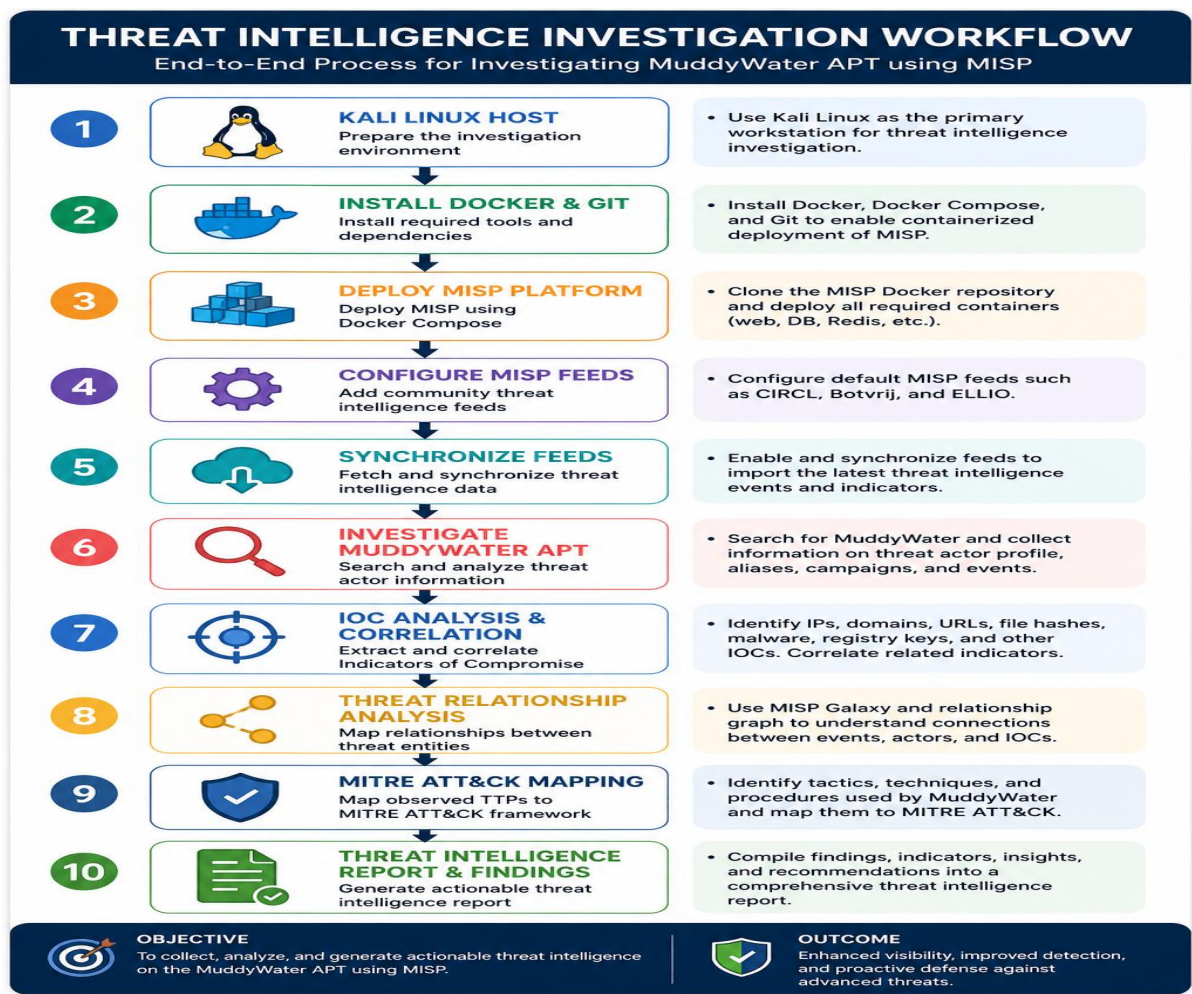


Figure 1. Threat Intelligence Investigation Workflow

3. Introduction and Organizational Background

3.1 Introduction

The rapid evolution of cyber threats has significantly increased the complexity of defending modern digital environments. Organizations operating within government, finance, healthcare, energy, and other critical infrastructure sectors continue to face persistent attacks from highly organized Advanced Persistent Threat (APT) groups capable of conducting long-term cyber espionage, credential theft, and disruptive cyber operations. Unlike opportunistic cybercriminals, APT groups possess substantial technical capabilities, strategic objectives, and the resources required to maintain prolonged access to targeted environments while remaining undetected.

As cyberattacks become increasingly sophisticated, organizations can no longer rely solely on traditional security controls such as firewalls, antivirus software, or signature-based detection systems. Modern cybersecurity operations require the integration of threat intelligence platforms capable of collecting, correlating, and sharing real-time intelligence from multiple trusted sources. This intelligence enables Security Operations Center (SOC) analysts to understand adversary behavior, identify Indicators of Compromise (IOCs), attribute malicious activities to known threat actors, and proactively defend organizational assets before attacks escalate into major security incidents.

The **Malware Information Sharing Platform (MISP)** has emerged as one of the most widely adopted open-source Threat Intelligence Platforms (TIPs), enabling organizations to share structured cyber threat intelligence, correlate malicious indicators across multiple data sources, and enrich investigations through collaborative intelligence sharing. By integrating community threat feeds, threat actor profiles, malware intelligence, and MITRE ATT&CK mappings, MISP provides security teams with valuable contextual information that supports faster detection, investigation, and incident response.

This project demonstrates how MISP can be deployed and utilized to investigate a real-world nation-state threat actor. The investigation focuses on **MuddyWater**, an Iranian state-aligned Advanced Persistent Threat (APT) group known for targeting government agencies, critical infrastructure, telecommunications providers, financial institutions, and defense organizations. Using threat intelligence available within MISP, the project analyzes the group's operational characteristics, extracts Indicators of Compromise (IOCs), examines associated threat campaigns, and maps observed adversary behaviors to the MITRE ATT&CK framework. The resulting intelligence provides actionable insights that strengthen organizational cyber defense capabilities and improve threat-informed decision-making.

3.2 Organizational Background

SecureOps Cybersecurity Inc. is a globally recognized cybersecurity organization headquartered in Europe, providing advanced cybersecurity services to government agencies, multinational corporations, financial institutions, healthcare organizations, and critical infrastructure providers. Since its establishment in 2010, the company has focused on delivering comprehensive security solutions that help clients identify, prevent, detect, investigate, and respond to evolving cyber threats.

The organization's core service offerings include:

- Threat Intelligence
- Security Operations Center (SOC) Services
- Incident Response and Digital Forensics
- Vulnerability Assessment and Risk Management
- Threat Hunting
- Security Monitoring
- Cyber Risk Advisory

To support these services, SecureOps operates a proprietary Threat Intelligence Platform (TIP) that integrates multiple intelligence sources, vulnerability assessments, attack surface monitoring, and real-time threat feeds. This intelligence-driven approach enables analysts to identify emerging threats, correlate malicious activity across diverse environments, and deliver timely security recommendations to clients.

One of SecureOps' primary competitive advantages lies in its ability to integrate **MISP** into its threat intelligence workflows. Through automated ingestion of trusted community intelligence feeds and collaborative information sharing, the organization enriches its internal threat database with high-confidence indicators, threat actor intelligence, malware information, and adversary tactics. This capability significantly improves threat visibility, accelerates incident investigations, and enhances proactive defense against sophisticated cyber threats.

As attacks against critical infrastructure continue to increase in both frequency and sophistication, SecureOps remains committed to strengthening its threat intelligence capabilities through continuous research, intelligence sharing, and the adoption of advanced cybersecurity technologies. This project forms part of that strategic initiative by demonstrating how structured threat intelligence can be leveraged to investigate one of today's most active nation-state adversaries.

Figure 2. MISP Project Directory and Repository Initialization

This figure illustrates the initial project workspace where the MISP investigation environment was organized prior to deployment.

```
(kali@kali)-[~]
$ mkdir MuddyWater-MISP-Threat-Intelligence
cd MuddyWater-MISP-Threat-Intelligence

(kali@kali)-[~/MuddyWater-MISP-Threat-Intelligence]
$ cd ~/MuddyWater-MISP-Threat-Intelligence

(kali@kali)-[~/MuddyWater-MISP-Threat-Intelligence]
$ pwd
/home/kali/MuddyWater-MISP-Threat-Intelligence

(kali@kali)-[~/MuddyWater-MISP-Threat-Intelligence]
$ git clone https://github.com/MISP/misp-docker
Cloning into 'misp-docker' ...
remote: Enumerating objects: 3256, done.
remote: Counting objects: 100% (57/57), done.
remote: Compressing objects: 100% (25/25), done.
remote: Total 3256 (delta 44), reused 35 (delta 32), pack-reused 3199 (from 2)
Receiving objects: 100% (3256/3256), 957.54 KiB | 335.00 KiB/s, done.
Resolving deltas: 100% (1788/1788), done.

(kali@kali)-[~/MuddyWater-MISP-Threat-Intelligence]
$ ls
misp-docker

(kali@kali)-[~/MuddyWater-MISP-Threat-Intelligence]
$
```

Figure 3. MISP Docker Repository Successfully Cloned

```
(kali@kali)-[~/MuddyWater-MISP-Threat-Intelligence/misp-docker]
$ cp template.env .env

(kali@kali)-[~/MuddyWater-MISP-Threat-Intelligence/misp-docker]
$ ls -la
total 176
drwxrwxr-x 10 kali kali 4096 Jul 20 14:49 .
drwxrwxr-x 3 kali kali 4096 Jul 20 14:46 ..
drwxrwxr-x 3 kali kali 4096 Jul 20 14:46 core
-rw-rw-r-- 1 kali kali 5265 Jul 20 14:46 docker-bake.hcl
-rw-rw-r-- 1 kali kali 17560 Jul 20 14:46 docker-compose.yml
drwxrwxr-x 3 kali kali 4096 Jul 20 14:46 docs
-rw-rw-r-- 1 kali kali 13658 Jul 20 14:49 .env
drwxrwxr-x 3 kali kali 4096 Jul 20 14:46 experimental
drwxrwxr-x 7 kali kali 4096 Jul 20 14:46 .git
drwxrwxr-x 3 kali kali 4096 Jul 20 14:46 .github
-rw-rw-r-- 1 kali kali 144 Jul 20 14:46 .gitignore
drwxrwxr-x 3 kali kali 4096 Jul 20 14:46 guard
drwxrwxr-x 5 kali kali 4096 Jul 20 14:46 kubernetes
-rw-rw-r-- 1 kali kali 35149 Jul 20 14:46 LICENSE
-rw-rw-r-- 1 kali kali 1479 Jul 20 14:46 Makefile
drwxrwxr-x 2 kali kali 4096 Jul 20 14:46 modules
-rw-rw-r-- 1 kali kali 30532 Jul 20 14:46 README.md
-rw-rw-r-- 1 kali kali 13658 Jul 20 14:46 template.env

(kali@kali)-[~/MuddyWater-MISP-Threat-Intelligence/misp-docker]
$ sudo docker compose pull
[sudo] password for kali:
WARN[0000] The "GUARD_ARGS" variable is not set. Defaulting to a blank string.
WARN[0000] The "GUARD_COMMIT" variable is not set. Defaulting to a blank string.
WARN[0000] The "SES_USER" variable is not set. Defaulting to a blank string.
WARN[0000] The "SES_PASSWORD" variable is not set. Defaulting to a blank string.
WARN[0000] The "SES_REGION" variable is not set. Defaulting to a blank string.
WARN[0000] The "SES_PORT" variable is not set. Defaulting to a blank string.
WARN[0000] The "CORE_COMMIT" variable is not set. Defaulting to a blank string.
WARN[0000] The "PYPI_JSONSCHEMA_VERSION" variable is not set. Defaulting to a blank string.
WARN[0000] The "PYPI_LIEF_VERSION" variable is not set. Defaulting to a blank string.
```

This figure shows the successful cloning of the official MISP Docker repository, providing the deployment files required to build the containerized threat intelligence platform.

Figure 4. MISP Docker Project Structure

This figure presents the project directory structure after cloning the repository, highlighting the configuration and deployment files used throughout the implementation.

```
(kali㉿kali)-[~/MuddyWater-MISP-Threat-Intelligence]
$ cd misp-docker

(kali㉿kali)-[~/MuddyWater-MISP-Threat-Intelligence/misp-docker]
$ pwd
/home/kali/MuddyWater-MISP-Threat-Intelligence/misp-docker

(kali㉿kali)-[~/MuddyWater-MISP-Threat-Intelligence/misp-docker]
$ ls -la
total 160
drwxrwxr-x 10 kali kali 4096 Jul 20 14:46 .
drwxrwxr-x  3 kali kali 4096 Jul 20 14:46 ..
drwxrwxr-x  3 kali kali 4096 Jul 20 14:46 core
-rw-rw-r--  1 kali kali 5265 Jul 20 14:46 docker-bake.hcl
-rw-rw-r--  1 kali kali 17560 Jul 20 14:46 docker-compose.yml
drwxrwxr-x  3 kali kali 4096 Jul 20 14:46 docs
drwxrwxr-x  3 kali kali 4096 Jul 20 14:46 experimental
drwxrwxr-x  7 kali kali 4096 Jul 20 14:46 .git
drwxrwxr-x  3 kali kali 4096 Jul 20 14:46 .github
-rw-rw-r--  1 kali kali  144 Jul 20 14:46 .gitignore
drwxrwxr-x  3 kali kali 4096 Jul 20 14:46 guard
drwxrwxr-x  5 kali kali 4096 Jul 20 14:46 kubernetes
-rw-rw-r--  1 kali kali 35149 Jul 20 14:46 LICENSE
-rw-rw-r--  1 kali kali  1479 Jul 20 14:46 Makefile
drwxrwxr-x  2 kali kali 4096 Jul 20 14:46 modules
-rw-rw-r--  1 kali kali 30532 Jul 20 14:46 README.md
-rw-rw-r--  1 kali kali 13658 Jul 20 14:46 template.env
```

4. Business Challenge, Project Rationale, Objectives, and Scope

4.1 Business Challenge

The cybersecurity threat landscape continues to evolve at an unprecedented pace, with nation-state Advanced Persistent Threat (APT) groups employing increasingly sophisticated techniques to compromise government agencies, critical infrastructure, financial institutions, healthcare organizations, and multinational enterprises. These adversaries leverage highly targeted phishing campaigns, custom malware, living-off-the-land techniques, credential theft, and stealthy command-and-control infrastructures to maintain long-term access to victim environments while evading traditional security controls.

SecureOps Cybersecurity Inc. has recently observed a significant increase in intelligence reports involving cyber espionage campaigns targeting several of its European clients. These incidents exhibit characteristics commonly associated with nation-state threat actors, including carefully crafted spear-phishing attacks, malicious document delivery, PowerShell-based execution, remote access malware, and extensive post-compromise reconnaissance activities. Despite the availability of multiple intelligence sources, analysts faced considerable challenges in accurately attributing these attacks to a specific threat actor due to the diversity of attack vectors, constantly evolving infrastructure, and fragmented threat intelligence.

The inability to rapidly correlate threat indicators across multiple intelligence sources resulted in delayed threat attribution, slower incident response, and reduced confidence in the intelligence products delivered to clients. This limitation affected SecureOps' ability to proactively advise customers on emerging threats and implement preventive security controls before attacks could progress further within targeted environments.

To address these challenges, SecureOps initiated a dedicated threat intelligence investigation utilizing the **Malware Information Sharing Platform (MISP)**. By integrating structured threat intelligence feeds, correlating Indicators of Compromise (IOCs), and leveraging collaborative threat intelligence, the organization sought to improve attribution accuracy, accelerate intelligence production, and strengthen proactive defense capabilities against sophisticated nation-state adversaries.

4.2 Project Rationale

Modern Security Operations Centers (SOCs) require more than reactive detection capabilities; they depend on accurate, timely, and contextual threat intelligence to anticipate adversary behavior and reduce organizational risk. Threat Intelligence Platforms such as **MISP** enable security teams to consolidate intelligence from trusted sources, correlate indicators across multiple campaigns, enrich investigations with contextual information, and share intelligence efficiently within and across organizations.

This project was initiated to demonstrate how MISP can be leveraged as a centralized threat intelligence platform for investigating a real-world Advanced Persistent Threat. By focusing on the MuddyWater threat actor, the project provides practical insight into the complete threat intelligence lifecycle—from intelligence collection and correlation to threat actor profiling and actionable reporting.

The investigation supports SecureOps' broader objective of enhancing its intelligence-driven security operations by improving threat visibility, enabling earlier detection of malicious activities, and strengthening the organization's ability to provide clients with high-confidence intelligence products that support informed cybersecurity decision-making.

Furthermore, this project serves as a reusable framework for future threat intelligence investigations involving other nation-state actors, ransomware groups, and emerging cyber threats.

4.3 Project Objectives

The primary objective of this project was to deploy and utilize the Malware Information Sharing Platform (MISP) to investigate, profile, and analyze the **MuddyWater Advanced Persistent Threat (APT)** group using structured cyber threat intelligence.

The investigation sought to transform raw threat intelligence into actionable information that could support Security Operations Center (SOC) operations, incident response activities, and strategic cybersecurity decision-making.

The specific objectives of the project were to:

- Deploy MISP using a Docker-based containerized environment on Kali Linux.
- Configure and synchronize trusted community threat intelligence feeds within MISP.
- Collect and analyze Indicators of Compromise (IOCs) associated with the MuddyWater threat actor.
- Develop a comprehensive threat actor profile, including aliases, motivations, geographic attribution, targeted sectors, tools, malware families, and operational characteristics.
- Examine correlated threat intelligence events and identify relationships between campaigns and malicious infrastructure.
- Map observed adversary behaviors to the **MITRE ATT&CK** framework to better understand the tactics, techniques, and procedures (TTPs) employed by the threat actor.
- Produce actionable threat intelligence capable of improving detection engineering, threat hunting, incident response, and defensive security operations.
- Develop a structured knowledge base that can support future investigations involving similar nation-state threat actors.

4.4 Project Scope

The scope of this project was limited to the collection, correlation, analysis, and reporting of threat intelligence associated with the **MuddyWater** threat actor using the Malware Information Sharing Platform (MISP).

The project focused specifically on:

- Deploying and configuring MISP within a Docker environment.
- Importing and synchronizing community threat intelligence feeds.
- Investigating MuddyWater-related threat intelligence events.

- Extracting Indicators of Compromise (IOCs), including malicious IP addresses, domains, URLs, registry artifacts, and cryptographic file hashes where available.
- Analyzing threat actor relationships using MISP Galaxy clusters.
- Examining known aliases, attribution, operational objectives, and targeted industries.
- Mapping identified Tactics, Techniques, and Procedures (TTPs) to the MITRE ATT&CK framework.
- Producing a comprehensive threat intelligence report containing findings, defensive recommendations, and strategic insights.

Activities outside the scope of this project include live malware reverse engineering, digital forensics on compromised systems, active threat hunting within production environments, penetration testing, and incident remediation. Instead, the project emphasizes intelligence-driven analysis using existing threat intelligence sources available within MISP.

Figure 5. MISP Core Services Successfully Started

```
WARN[0000] The "COMPOSE_PROFILES" variable is not set. Defaulting to a blank
[+] up 4/4
✓ Container misp-docker-db-1           Healthy
✓ Container misp-docker-misp-modules-1 Healthy
✓ Container misp-docker-redis-1        Healthy
✓ Container misp-docker-misp-core-1    Started

(kali@kali) - [~/MuddyWater-MISP-Threat-Intelligence/misp-docker]
```

This figure demonstrates the successful initialization of the core MISP services within the Docker environment, confirming that the platform was ready for configuration and threat intelligence operations.

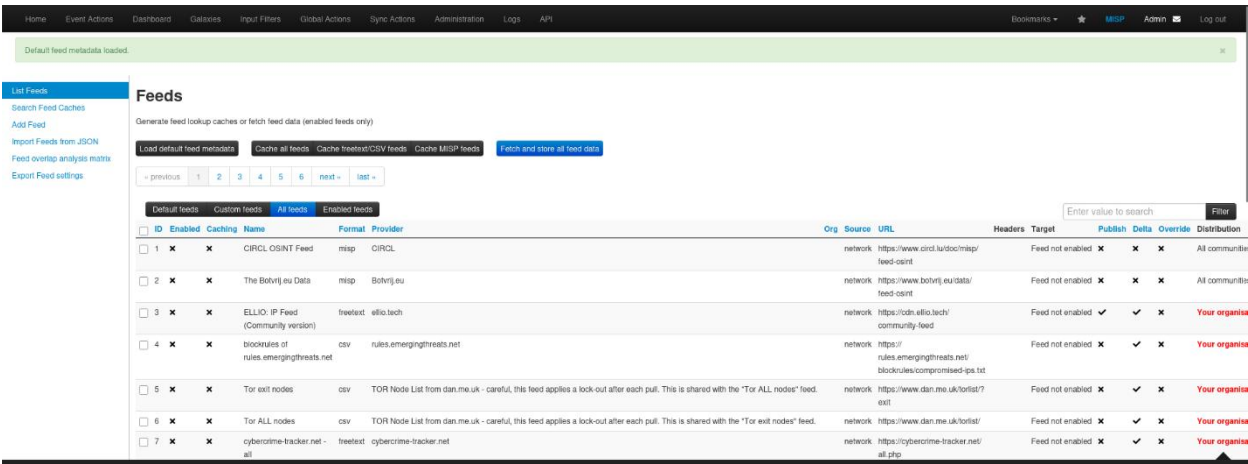
Figure 6. MISP Docker Stack Running Successfully

This figure verifies the operational status of the complete MISP Docker stack, indicating that all required containers were running correctly prior to beginning the threat intelligence investigation.

```
WARN[0000] The "SES_PORT" variable is not set. Defaulting to a blank string.
NAME                IMAGE                COMMAND                SERVICE    CREATED      STATUS        PORTS
misp-docker-db-1    mariadb:10.11        "docker-entrypoint.s..." db         10 minutes ago Up 10 minutes (healthy) 3306/tcp
misp-docker-mail-1  ghcr.io/egos-tech/smt... "/bin/entrypoint.sh ..." mail       10 minutes ago Up 10 minutes          25/tcp
misp-docker-misp-core-1  ghcr.io/misp/misp-docker/misp-core:latest "/entrypoint.sh"      misp-core  10 minutes ago Up 34 seconds (healthy) 0.0.0.0:80→80/tcp, [::]:80→80/tcp, 0.0.0.0:443→443/tcp, [::]:443→443/tcp
misp-docker-misp-modules-1  ghcr.io/misp/misp-docker/misp-modules:latest "/usr/local/bin/misp..." misp-modules 10 minutes ago Up 10 minutes (healthy)
misp-docker-redis-1  valkey/valkey:7.2    "docker-entrypoint.s..." redis      10 minutes ago Up 10 minutes (healthy) 6379/tcp

(kali@kali) - [~/MuddyWater-MISP-Threat-Intelligence/misp-docker]
```

Figure 7. MISP User Authentication



This figure shows successful authentication into the MISP web interface, providing access to the platform's threat intelligence management and analysis capabilities.

5. Project Methodology and Lab Environment

5.1 Project Methodology

A structured threat intelligence methodology was adopted to ensure that intelligence collection, analysis, and reporting followed a logical and repeatable process. Rather than simply collecting Indicators of Compromise (IOCs), the investigation focused on transforming raw threat data into actionable intelligence that could support Security Operations Center (SOC) operations, incident response, and strategic decision-making.

The investigation followed the Threat Intelligence Lifecycle, comprising the following phases:

Phase 1 – Environment Preparation

The project began with the preparation of a dedicated threat intelligence environment using **Kali Linux** as the primary operating system. Required software packages, including Docker, Docker Compose, and Git, were installed to support deployment of the Malware Information Sharing Platform (MISP).

Phase 2 – Platform Deployment

The official MISP Docker repository was cloned, and the platform was deployed using Docker Compose. Containerization provided an isolated, portable, and scalable deployment environment while simplifying installation and dependency management.

Once deployment was complete, all required services—including the web server, database, Redis, and supporting containers—were verified to ensure the platform was fully operational before intelligence collection commenced.

Phase 3 – Threat Intelligence Feed Configuration

After successfully accessing the MISP web interface, community threat intelligence feeds were configured.

The investigation utilized selected default MISP feeds, specifically:

- CIRCL OSINT Feed
- Botvrij Threat Feed
- ELLIO Threat Feed

These feeds were enabled, cached, and synchronized to import current threat intelligence into the local MISP instance.

The imported intelligence provided access to threat events, malicious infrastructure, malware information, Indicators of Compromise (IOCs), and threat actor profiles that formed the foundation of the investigation.

Phase 4 – Threat Actor Investigation

Using MISP's search and correlation capabilities, the investigation focused on the **MuddyWater** Advanced Persistent Threat group.

Threat intelligence analysis included:

- Threat actor profiling
- Event correlation
- IOC extraction
- Attribution analysis
- Galaxy cluster investigation
- Relationship mapping
- MITRE ATT&CK analysis

Multiple threat intelligence objects were examined to understand the adversary's operational characteristics, infrastructure, and historical campaigns.

Phase 5 – Intelligence Analysis

Collected threat intelligence was analyzed to identify meaningful relationships between indicators.

Particular attention was given to:

- Known malicious IP addresses
- Domains
- URLs
- File hashes
- Malware families
- Registry artifacts
- Threat actor aliases
- Campaign relationships

The investigation also assessed how these indicators could be operationalized within Security Operations Center (SOC) environments to improve threat detection and incident response.

Phase 6 – Intelligence Reporting

The final phase involved documenting all findings in a structured threat intelligence report.

The report consolidates:

- Technical findings
- Threat actor profile
- Indicators of Compromise
- MITRE ATT&CK mappings
- Threat relationships
- Security recommendations
- Strategic defensive guidance

The resulting document serves as both a technical reference and an intelligence product that can support future investigations.

5.2 Lab Environment

To ensure consistency, reproducibility, and isolation from production systems, the investigation was conducted within a dedicated virtual laboratory.

The environment consisted of open-source technologies commonly used by Threat Intelligence and Security Operations Center (SOC) teams.

Operating System

- Kali Linux

Kali Linux served as the primary investigation workstation due to its comprehensive collection of cybersecurity tools and compatibility with Docker-based deployments.

Threat Intelligence Platform

- Malware Information Sharing Platform (MISP)

MISP functioned as the central Threat Intelligence Platform (TIP), enabling intelligence sharing, IOC management, event correlation, threat actor profiling, and MITRE ATT&CK integration.

Containerization

- Docker
- Docker Compose

Docker was used to deploy MISP and its supporting services within isolated containers, ensuring a standardized and portable deployment while reducing installation complexity.

Version Control

- Git

Git was used to clone the official MISP Docker repository and manage project files throughout the implementation.

Threat Intelligence Sources

The investigation leveraged trusted community intelligence feeds available within MISP.

Configured feeds included:

- CIRCL OSINT Feed
- Botvrij Threat Feed
- ELLIO Threat Feed

These feeds continuously provide structured cyber threat intelligence that includes malware indicators, threat actor intelligence, attack infrastructure, vulnerabilities, and associated threat events.

Threat Intelligence Framework

The **MITRE ATT&CK Framework** was used to classify and understand adversary behavior by mapping observed Tactics, Techniques, and Procedures (TTPs) to a standardized knowledge base. This mapping enhances the ability of defenders to develop detection logic, conduct threat hunting, and prioritize defensive measures based on known adversary tradecraft.

5.3 Technical Workflow

The implementation followed the workflow illustrated below:

Kali Linux

|

▼

Install Docker & Git

|

▼

Clone MISP Docker Repository

|

▼

Deploy MISP Containers

|

▼

Verify Docker Services

|

▼

Login to MISP

|

▼

Load Default Feed Metadata

|



Enable Threat Feeds

|



Cache & Fetch Threat Intelligence

|



Search MuddyWater

|



Threat Actor Profiling

|



IOC Extraction

|



MITRE ATT&CK Mapping

|



Threat Relationship Analysis

|



Threat Intelligence Report

This workflow demonstrates the end-to-end process followed throughout the investigation, from platform deployment to the production of actionable threat intelligence.

Home
Event Actions
Dashboard
Galleries
Input Fields
Global Actions
Synic Actions
Administration
Logs
API

[Bookmark](#)

[MISP](#)
[Admin](#)
[Log out](#)

List Events

Add Event

Import from ...

REST client

List Attributes

Search Attributes

View Proposals

Events with proposals

View delegation requests

View periodic summary

Export

Automation

Events

- previous 1 2 3 4 5 6 7 8

☐		Creator org	Owner org	ID	Clusters
☐				79	Enterprise Attack Mimikatz - 500 PsExec - 5002
☐				234	Enterprise Attack Mimikatz - 500 PsExec - 5002
☐				457	Microsoft Active Directory Mango Sandstorm Enterprise Attack MuddyWater - Intrusion Set MuddyWater - Threat Actor MuddyWater - Attack Pattern Spearphishing Country Israel
☐				454	Attack Pattern OS Credential Data from Logs Application ID Device ID

MuddyWater

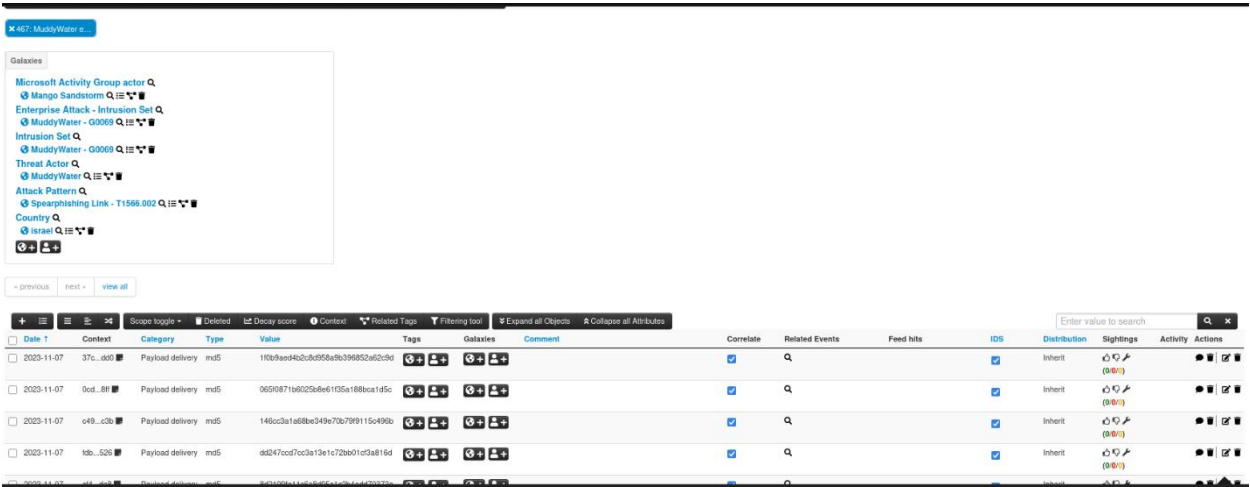
Description: The MuddyWater attacks are primarily against Middle Eastern nations. However, we have also observed attacks against surrounding nations and beyond, including targets in India and the USA. MuddyWater attacks are characterized by the use of a slowly evolving PowerShell-based first stage backdoor we call "POWERSTATS". Despite broad scrutiny and reports on MuddyWater attacks, the activity continues with only incremental changes to the tools and techniques.

Synonyms:
 TEMP Zagros
 Static Kitten
 Sledworm
 MERCURY
 COBALT ULLSTER
 QOO99
 ATKSI
 Boggy Serpens
 Mango Sandstorm
 TA450
 Earth Vitalia
 Source: MISP Project
 Attribution Confidence: 50
 CTR Suspected State Sponsor: Iran (Islamic Republic of)
 CTR Suspected Victims:
 Saudi Arabia
 Georgia
 Turkey
 Iraq
 Israel
 India
 United Arab Emirates

#Attr.	#Corr.	Creator user	Date	Info
635		admin@admin.test	2026-07-20	ELLIO: IP Feed (Com
1299		admin@admin.test	2020-09-08	Banco Estado Chile R
29		admin@admin.test	2023-11-07	MuddyWater vN-Ebi
14		admin@admin.test	2023-11-02	Windows version of f

MuddyWater eN-Abile spear-phishing with new TTPs	
Event ID	467
UUID	43ea5673-3bd8-4996-8d2f-66e56675914  
Creator org	CUDESO
Owner org	 ADMIN
Creator user	admin@admin.test
Protected Event (experimental) 	 Event is in unprotected mode.
Tags	 tip:white    
Date	2023-11-07
Threat Level	 Medium
Analysis	Completed
Distribution	All communities  
Published	 Yes First published at: 2026-07-20 14:48:25 Last published at: 2026-07-20 14:48:25
#Attributes	29 (0 Objects)
First recorded change	2023-11-07 17:30:51
Last change	2023-11-07 17:36:02
Modification map	
Sightings	0 (0) - restricted to own organisation only. 

Figure 10. Threat Intelligence Events Successfully Imported into MISP



This figure confirms the successful synchronization and import of threat intelligence events into the local MISP instance, providing the dataset used for the MuddyWater investigation.

6. Threat Intelligence Investigation and Analysis

6.1 Investigation Overview

Following the successful deployment and configuration of the Malware Information Sharing Platform (MISP), the investigation transitioned into the intelligence analysis phase. This phase focused on identifying, analyzing, and profiling the **MuddyWater Advanced Persistent Threat (APT)** using intelligence gathered from synchronized community threat feeds.

The investigation aimed to answer several key intelligence questions:

- Who is the threat actor?
- What are the threat actor's motivations?
- Which industries are being targeted?
- What Indicators of Compromise (IOCs) are associated with the threat actor?
- What tactics, techniques, and procedures (TTPs) are commonly employed?
- Which malware families and tools are linked to the threat actor?
- How can this intelligence improve proactive cyber defense?

By leveraging MISP's threat intelligence correlation capabilities, analysts were able to consolidate multiple intelligence sources into a unified view of the adversary. This enabled

SecureOps Cybersecurity Inc. to move beyond isolated indicators and develop a comprehensive understanding of the threat actor's operational behavior.

6.2 Threat Actor Identification

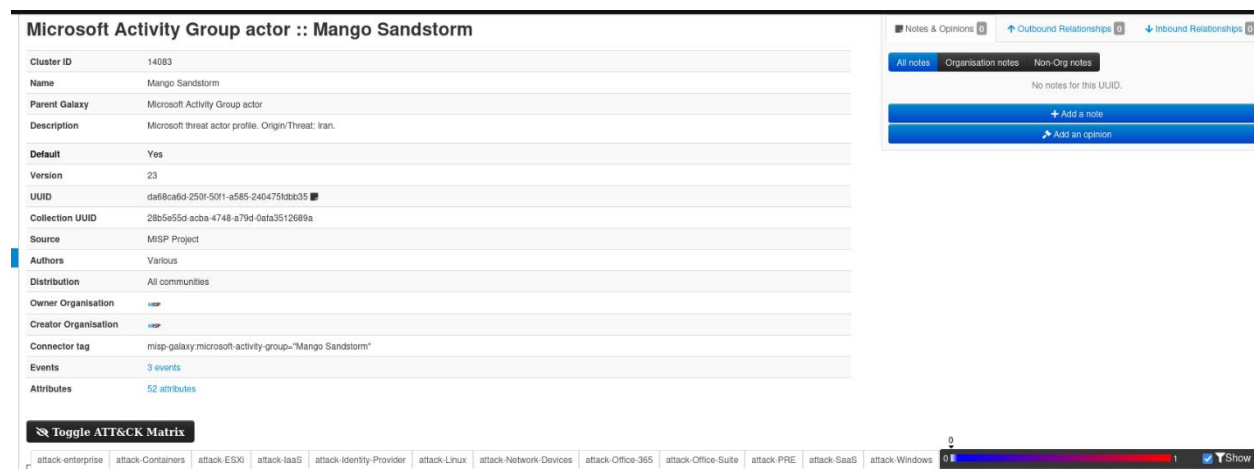
The investigation identified **MuddyWater** as the primary threat actor associated with the imported intelligence events.

MuddyWater is a highly active **Advanced Persistent Threat (APT)** group widely assessed by multiple cybersecurity organizations to be aligned with the Iranian government. The group has been active since approximately **2017** and is primarily associated with cyber espionage campaigns targeting government agencies, critical infrastructure operators, telecommunications providers, financial institutions, defense organizations, and energy companies across Europe, the Middle East, North America, and Asia.

Unlike financially motivated cybercriminal groups, MuddyWater's operations primarily focus on intelligence gathering, long-term persistence within victim networks, credential theft, surveillance, and strategic information collection. The group is known for adapting its infrastructure, changing malware families, and frequently using legitimate system administration tools to blend into normal system activity and evade detection.

The threat actor has consistently demonstrated a preference for **Living-off-the-Land (LOLBin)** techniques, abusing legitimate Windows utilities such as PowerShell and other native administrative tools instead of relying exclusively on custom malware. This operational approach significantly complicates detection efforts because many of the executed commands resemble legitimate administrative activities.

Figure 11. MuddyWater Event Overview



This figure displays the MuddyWater event within MISP, providing analysts with an overview of the associated threat intelligence event, event metadata, and linked intelligence objects used throughout the investigation.

6.3 Threat Actor Profile

Threat actor profiling is a critical component of cyber threat intelligence because it enables analysts to understand an adversary's identity, objectives, capabilities, operational methods, and historical activities. Rather than focusing solely on individual Indicators of Compromise (IOCs), profiling provides strategic context that supports attribution, risk assessment, threat hunting, and defensive planning.

The investigation revealed the following profile for MuddyWater.

Attribute	Details
Threat Actor	MuddyWater
Threat Type	Advanced Persistent Threat (APT)
Country of Origin	Iran
Suspected Sponsor	Ministry of Intelligence and Security (MOIS)
Primary Motivation	Cyber Espionage and Strategic Intelligence Collection
Operational Goal	Long-term access to targeted environments for intelligence gathering
Active Since	Approximately 2017
Primary Targets	Government, Defense, Energy, Telecommunications, Financial Services, Healthcare, Critical Infrastructure
Geographic Focus	Europe, Middle East, North America, Asia

The intelligence collected through MISP indicates that MuddyWater continuously evolves its operational infrastructure and attack methodologies. Rather than relying on a single malware family or command-and-control infrastructure, the group frequently modifies its operational toolkit, making attribution more difficult without correlated threat intelligence.

This adaptability underscores the importance of platforms such as MISP, where analysts can correlate indicators, threat actor profiles, and historical campaigns to reveal patterns that may not be evident when analyzing isolated security events.

Figure 12. Microsoft Threat Actor Profile (Mango Sandstorm)

Tidal Groups :: MuddyWater	
Cluster ID	38130
Name	MuddyWater
Parent Galaxy	Tidal Groups
Description	MuddyWater is a cyber espionage group assessed to be a subordinate element within Iran's Ministry of Intelligence and Security (MOIS).^[CYBERCOM Iranian Intel Cyber January 2022] Since at least 2017, MuddyWater has targeted a range of government and private organizations across sectors, including telecommunications, local government, defense, and oil and natural gas organizations, in the Middle East, Asia, Africa, Europe, and North America.^[Unit 42 MuddyWater Nov 2017]^[Symantec MuddyWater Dec 2018]^[ClearSky MuddyWater Nov 2018]^[ClearSky MuddyWater June 2019]^[Reaqta MuddyWater November 2017]^[DHS CISA AA22-055A MuddyWater February 2022]^[Talos MuddyWater Jan 2022]
Default	Yes
Version	1
UUID	dcb260d8-9d53-404f-9ff5-dbee2c6ef6e6
Collection UUID	877cdc4b-3392-4353-a7d4-2e46d40e5936
Source	https://app-api.tidalcyber.com/api/v1/groups/
Authors	Tidal Cyber
Distribution	All communities
Owner Organisation	MISP
Creator Organisation	MISP
Connector tag	misp-galaxy:groups="MuddyWater"
Events	1 event
Attributes	11 attributes

This figure presents Microsoft's intelligence profile for Mango Sandstorm, one of the primary names used by Microsoft to track the MuddyWater threat actor. The profile provides additional context regarding the group's objectives, operational history, and observed activity.

Figure 13. MuddyWater Threat Actor Profile

Attributes

previous

next

Search

Date	Event	Org	Category	Type	Value	Tags	Galaxies	Comment	Correlate	Related Events	Feed hits	IDS	Distribution	Sightings	Activity	Actions
2016-02-21	1	CUDESO	Artifacts dropped	sha1	af364ff503da71875b6d7c401a1e98e31450a561					Q			Inherit event			
2016-02-21	1	CUDESO	Network activity	ip-dst	84.11.146.62					Q			Inherit event			
2016-02-21	1	CUDESO	Network activity	ip-dst	107.6.172.54					Q			Inherit event			
2016-02-21	1	CUDESO	Network activity	ip-dst	107.6.181.116					Q			Inherit event			
2016-02-21	1	CUDESO	Artifacts dropped	sha1	dc2b6f40b4e76c2f538a3a8e22c35547c8020af					Q			Inherit event			
2016-02-21	1	CUDESO	Artifacts dropped	sha1	29968b0c415722676107333f2a82b588d8f8e					Q			Inherit event			
2016-02-21	1	CUDESO	Artifacts dropped	sha1	eeb67e663b2f980cd8c228f2e04304c8992401d					Q			Inherit event			
2016-02-21	1	CUDESO	Artifacts dropped	sha1	c80960780f6c7bb6482c5b00211802168f9c8a					Q			Inherit event			
2016-02-21	1	CUDESO	Artifacts dropped	sha1	45754e9a0322648f95a8e339d9d9a023fa7					Q			Inherit event			
2016-02-21	1	CUDESO	Artifacts dropped	sha1	46a995d8d9918ca0793404110904479b6adcb9f					Q			Inherit event			
2016-02-21	1	CUDESO	Network activity	url	http://gulic.halta.ac.il/index.php/the-ezri-center-in-the-media/291-the-ezri-center-in-the-media					Q			Inherit event			
2016-02-21	1	CUDESO	Network activity	domain	gulic.halta.ac.il					Q			Inherit event			
2016-06-21	1	CUDESO	Artifacts dropped	registry	HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run					1458 949 778 28 Q			Inherit event			
2016-02-21	1	CUDESO	Artifacts dropped	sha1	64ba130e027d485c5d50534e769d239000e068dd					Q			Inherit event			

This figure displays the detailed MuddyWater threat actor profile available within MISP, including attribution, descriptive intelligence, and associated threat information.

6.4 Known Aliases and Threat Attribution

One of the challenges in threat intelligence is that different cybersecurity vendors often assign different names to the same threat actor. To improve intelligence sharing and avoid attribution errors, analysts must understand these naming conventions.

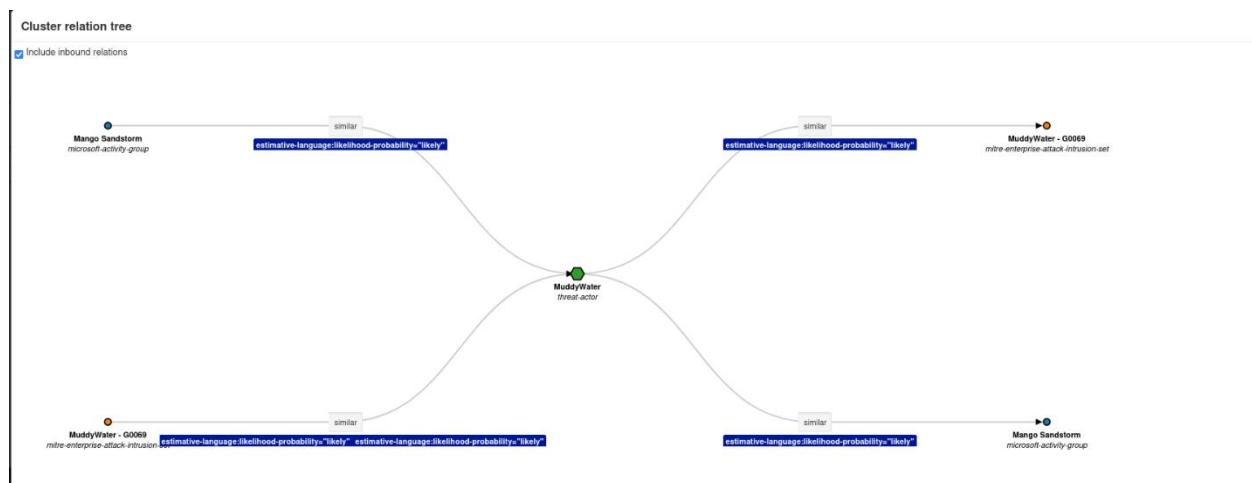
During the investigation, MISP correlated MuddyWater with several widely recognized aliases used by major cybersecurity organizations.

Organization	Threat Actor Name
Microsoft	Mango Sandstorm
CrowdStrike	Static Kitten
Secureworks	Seedworm
FireEye / Mandiant (historical reporting)	TEMP.Zagros
Proofpoint	TA450
SentinelOne	Boggy Serpens
Common Industry Name	MuddyWater

Although these names differ, they refer to the same or substantially overlapping threat activity attributed to the Iranian state-aligned espionage group. Understanding these aliases enables analysts to correlate intelligence across multiple vendors and improves attribution accuracy during investigations.

Threat attribution remains an evolving process and should be interpreted alongside technical evidence, intelligence confidence, and corroborating sources rather than relying solely on naming conventions.

Figure 14. MuddyWater Aliases and Attribution



This figure illustrates the various aliases associated with MuddyWater as represented within MISP, helping analysts correlate reporting from different cybersecurity vendors.

6. Threat Intelligence Investigation and Analysis (Part B)

6.5 Indicators of Compromise (IOCs)

Indicators of Compromise (IOCs) are forensic artifacts observed on a network or endpoint that provide evidence of malicious activity. These indicators enable Security Operations Center (SOC) analysts to detect, investigate, contain, and remediate cyber threats before they escalate into major security incidents.

During the investigation, MISP automatically correlated intelligence from multiple community threat feeds and presented a consolidated collection of IOCs associated with the MuddyWater threat actor. These indicators included malicious domains, IP addresses, URLs, registry artifacts, filenames, malware samples, and cryptographic file hashes that have been observed across previous MuddyWater campaigns.

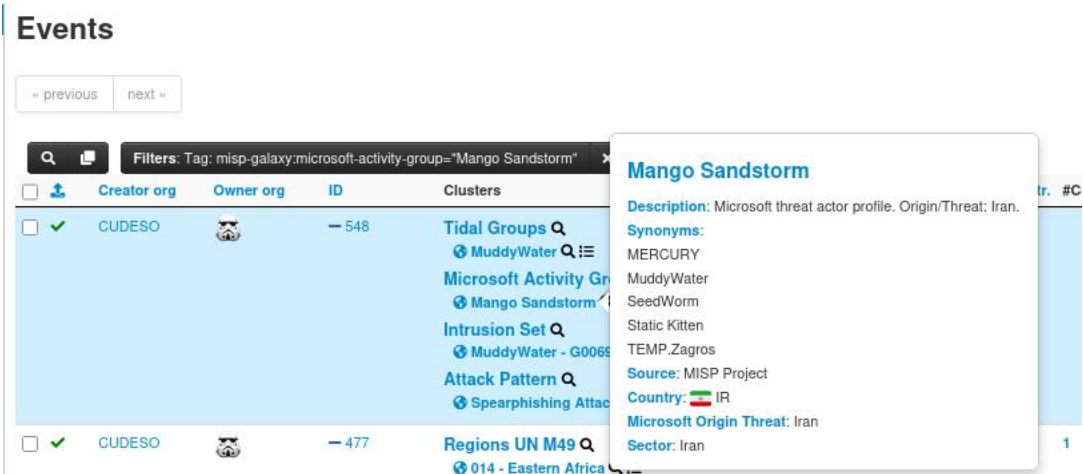
Rather than viewing each indicator in isolation, MISP correlated these artifacts with related threat events, malware families, and threat actor profiles, providing analysts with valuable context for threat attribution and operational decision-making.

The extracted IOCs serve multiple defensive purposes, including:

- Threat hunting across enterprise environments.
- SIEM detection rule creation.
- Firewall and IDS/IPS blocking.

- Endpoint Detection and Response (EDR) enrichment.
- Incident response investigations.
- Threat intelligence sharing.

Figure 15. MuddyWater Indicators of Compromise (IOCs)



This figure displays the Indicators of Compromise (IOCs) associated with the MuddyWater threat actor within MISP. The dataset includes multiple indicator types that support threat detection, correlation, and attribution.

6.5.1 Extracted Indicators of Compromise (IOCs)

Overview

During the threat intelligence investigation, the Malware Information Sharing Platform (MISP) identified multiple Indicators of Compromise (IOCs) associated with the MuddyWater Advanced Persistent Threat (APT). These indicators represent observable evidence of malicious activity and can be operationalized within Security Operations Centers (SOCs) to support threat detection, incident response, and threat hunting activities.

The extracted indicators include network-based, host-based, and file-based artifacts such as malicious IP addresses, domains, URLs, cryptographic hashes, filenames, registry artifacts, and malware references. These IOCs enable defenders to identify known malicious activity, correlate security events, and strengthen defensive controls across enterprise environments.

The table below summarizes the indicators extracted during the investigation.

Operational Value of the Extracted IOCs

The extracted indicators can be integrated into organizational security technologies to enhance detection and response capabilities.

Typical operational use cases include:

- SIEM alert enrichment and correlation.
- Firewall and web proxy blocklists.
- Endpoint Detection and Response (EDR) threat detection.
- Intrusion Detection and Prevention Systems (IDS/IPS).
- Threat hunting across enterprise endpoints and network traffic.
- Malware detection using cryptographic file hashes.
- Incident response investigations and forensic validation.

Because threat actors frequently rotate infrastructure and modify malicious artifacts, these indicators should be combined with behavioral detections and regularly updated through trusted threat intelligence feeds.

MuddyWater Indicators of Compromise (IOCs)

☐	Date	Context	Category	Type	Value	Tags	Galaxies	Comment	Correlate	Related Events	Feed hits	IDS	Distribution	Sightings	Activity	Actions
☐	2023-11-07	043_792	Payload delivery	md5	590753b308393c38010c8519c693c12					Q	2		Inherit	(0/0)		
☐	2023-11-07	4db_aaf	Network activity	url	ws.onehub.com/files/7f9dxtf6					Q	2		Inherit	(0/0)		
☐	2023-11-07	6e2_56e	Network activity	url	a.storyblok.com/1/253959/x/b92ea48421/form.zip					Q	2		Inherit	(0/0)		
☐	2023-11-07	415_24b	Network activity	url	a.storyblok.com/1/255988/x/5e0186161d/questionnaire.zip					Q	2		Inherit	(0/0)		
☐	2023-11-07	3c7_d5e	Network activity	url	a.storyblok.com/1/259791/x/94f59e378/questionnaire.zip					Q	2		Inherit	(0/0)		
☐	2023-11-07	d9e_da6	Network activity	ip-dst	146.70.149.61					Q	2		Inherit	(0/0)		
☐	2023-11-07	ada_87e	Network activity	ip-dst	146.70.124.102					477 Q	2		Inherit	(0/0)		
☐	2023-11-07	497_04a	Network activity	ip-dst	37.120.237.204					Q	2		Inherit	(0/0)		
☐	2023-11-07	b13_152	Network activity	ip-dst	37.120.237.248					Q	2		Inherit	(0/0)		
☐	2023-11-07	a9c_107	Network activity	url	a.storyblok.com/1/259837/x/21e6a04837/defense-video.zip					Q	2		Inherit	(0/0)		
☐	2023-11-07	167_566	Network activity	url	a.storyblok.com/1/259791/x/91e2f5a2f/attachments.zip					Q	2		Inherit	(0/0)		
☒	2023-11-07	277_316	Payload delivery	md5	16923d827a440161217b66a04e8b40a			Altera Agent Installer		Q	2		Inherit	(0/0)		

This figure presents the Indicators of Compromise (IOCs) extracted from the MuddyWater threat intelligence event within MISP. The displayed indicators include multiple IOC types used to support threat detection, event correlation, and adversary attribution.

S/N	IOC Type	IOC Value	Category	Description / Comment
1	URL	ws.onehub.com/files/7f9dxtt6	Network Activity	Malware delivery URL
2	URL	a.storyblok.com/f/253959/x/b92ea48421/form.zip	Network Activity	Malicious archive
3	URL	a.storyblok.com/f/255988/x/5e0186f61d/questionnaire.zip	Network Activity	Malicious questionnaire archive
4	URL	a.storyblok.com/f/259791/x/94f59e378f/questionnaire.zip	Network Activity	Malicious questionnaire archive
5	URL	a.storyblok.com/f/259837/x/21e6a04837/defense-video.zip	Network Activity	Malicious ZIP payload
6	URL	a.storyblok.com/f/259791/x/91e2f5fa2f/attachments.zip	Network Activity	Malicious attachment archive
7	IPv4 Address	146.70.149.61	Network Activity	Malicious destination IP
8	IPv4 Address	146.70.124.102	Network Activity	Malicious destination IP
9	IPv4 Address	37.120.237.204	Network Activity	Malicious destination IP
10	IPv4 Address	37.120.237.248	Network Activity	Malicious destination IP
11	MD5 Hash	16923d827a440161217fb66a04e8b40a	Payload Delivery	Atera Agent Installer
12	MD5 Hash	7568062ad4b22963f3930205d1a14df7	Payload Delivery	Archive containing Atera Agent

S/N	IOC Type	IOC Value	Category	Description / Comment
13	MD5 Hash	39eea24572c14910b67242a16e24b768	Payload Delivery	Archive containing Atera Agent
14	MD5 Hash	2e09e53135376258a03b7d793706b70f	Payload Delivery	Atera Agent Installer
15	MD5 Hash	37c3f5b3c814e2c014abc1210e8e69a2	Payload Delivery	Archive containing Atera Agent
16	MD5 Hash	1f0b9aed4b2c8d958a9b396852a62c9d	Payload Delivery	Malware sample
17	MD5 Hash	065f0871b6025b8e61f35a188bca1d5c	Payload Delivery	Malware sample
18	MD5 Hash	146cc3a1a68be349e70b79f9115c496b	Payload Delivery	Malware sample
19	MD5 Hash	dd247ccd7cc3a13e1c72bb01cf3a816d	Payload Delivery	Malware sample
20	MD5 Hash	8d2199fa11c6a8d95c1c2b4add70373a	Payload Delivery	Malware sample
21	MD5 Hash	04aff1465a223a806774104b652a4f0	Payload Delivery	Malware sample
22	MD5 Hash	6167f03c8b2734c20eb02d406d3ba651	Payload Delivery	Malware sample
23	MD5 Hash	e8f3ecc0456fcbbb029b1c27dc1faad0	Payload Delivery	Malware sample
24	MD5 Hash	952cc4e278051e349e870aa80babc755	Payload Delivery	Malware sample
25	MD5 Hash	ea026027f283e7e16389853642635fc0	Payload Delivery	Malware sample

S/N	IOC Type	IOC Value	Category	Description / Comment
26	MD5 Hash	08802238e1c7efac1c664b3e81bb2a85	Payload Delivery	Malware sample
27	MD5 Hash	eb66eecd7c9b6663e1c3636b4a23d376	Payload Delivery	Malware sample
28	MD5 Hash	590753b3b08393c380f0c8519c693c12	Payload Delivery	Malware sample
29	External Intelligence Link	https://www.deepinstinct.com/blog/muddywater-en-able-spear-phishing-with-new-ttps	External Analysis	Technical reference for the campaign

IOC Summary

IOC Category	Count
URLs	6
IPv4 Addresses	4
MD5 File Hashes	18
External Intelligence Links	1
Total Indicators	29

Analysis of the Extracted Indicators

The extracted Indicators of Compromise reveal that the observed MuddyWater campaign relied primarily on malicious ZIP archives, download URLs, and destination IP addresses to deliver malware and establish attacker-controlled communications. Several of the payloads were identified as Atera Agent Installer archives, indicating the threat actor's use of legitimate remote administration software to establish persistence and remote access within compromised environments.

The MD5 hashes provide unique fingerprints for malware identification and can be integrated into antivirus platforms, Endpoint Detection and Response (EDR) solutions, malware sandboxes, and Security Information and Event Management (SIEM) platforms to facilitate rapid detection

and incident response. Likewise, the identified URLs and IP addresses can be incorporated into web proxies, firewalls, DNS filtering solutions, and intrusion detection systems to block known malicious infrastructure.

6.5.2 IOC Categories Identified

The investigation identified several categories of Indicators of Compromise commonly associated with MuddyWater operations.

IOC Category	Purpose
IP Addresses	Command-and-Control (C2) infrastructure and attacker-controlled servers.
Domains	Malicious infrastructure used for phishing, payload delivery, and communications.
URLs	Payload hosting, credential harvesting, and malware distribution.
File Names	Malware executables, scripts, and dropped payloads.
Registry Keys	Windows persistence mechanisms used to survive system reboots.
Email Addresses	Phishing campaigns and attacker-controlled accounts.
File Hashes	Unique identifiers used to detect known malware samples.

These indicators enable defenders to identify both current and historical adversary activity while improving detection accuracy across multiple security technologies.

6.5.3 Cryptographic File Hashes

One of the most reliable Indicators of Compromise is the cryptographic hash of a malicious file. Unlike filenames, which can easily be changed by attackers, cryptographic hashes provide a unique fingerprint that allows security teams to identify known malware with a high degree of confidence.

MISP supports multiple hash algorithms, including:

- MD5
- SHA-1
- SHA-256
- SHA-512
- ssdeep (fuzzy hashing)

These values are widely used by antivirus vendors, EDR platforms, malware sandboxes, and digital forensic tools to identify known malicious files and correlate malware samples across investigations.

Note: The specific hash values documented in this report are those available within the imported MISP event. Additional hashes may become available as community threat intelligence feeds are updated.

6.5.4 Malicious Infrastructure

The investigation also identified infrastructure used by MuddyWater during various campaigns.

Observed infrastructure included:

- Internet Protocol (IP) addresses
- Fully Qualified Domain Names (FQDNs)
- Uniform Resource Locators (URLs)
- Command-and-Control (C2) servers

Threat actors frequently rotate infrastructure to evade detection. Therefore, defenders should avoid relying solely on static blocklists and instead combine IOC-based detection with behavioral analytics, network monitoring, and threat intelligence correlation.

6.6 Associated Threat Events

One of MISP's strongest capabilities is event correlation. Rather than storing isolated indicators, MISP groups related intelligence into events that represent observed threat activity, malware campaigns, vulnerability exploitation, or adversary operations.

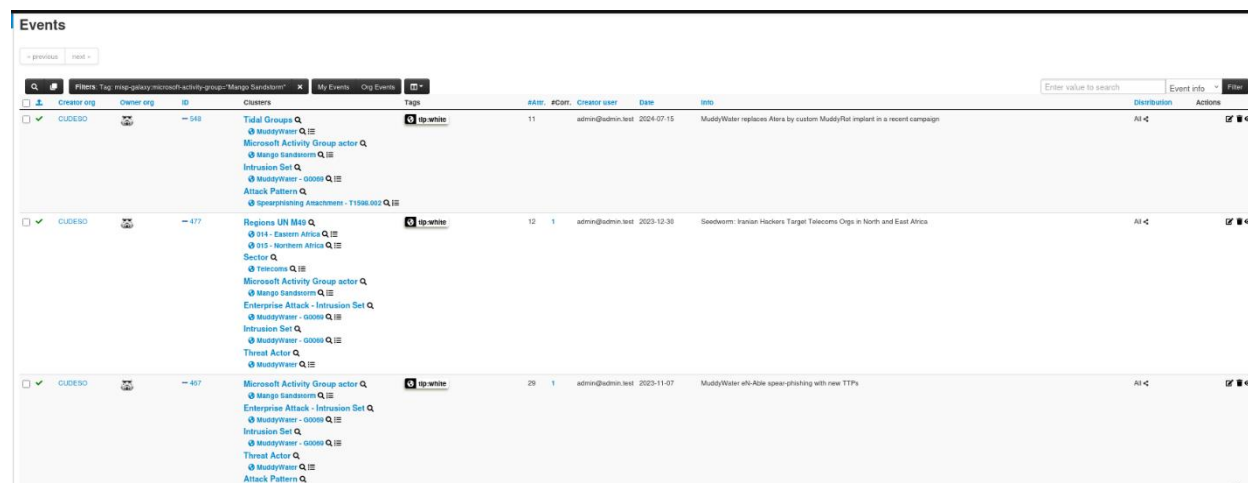
During the investigation, multiple events were associated with MuddyWater. These events provided contextual information such as:

- Campaign descriptions
- Attack timelines
- Malware associations
- Infrastructure reuse
- Intelligence sources
- Related Indicators of Compromise
- Threat actor relationships

Correlating these events allowed analysts to identify recurring operational patterns and understand how MuddyWater has adapted its infrastructure and techniques over time.

This contextual intelligence significantly improves attribution confidence and enables analysts to distinguish isolated malicious indicators from coordinated adversary campaigns.

Figure 16. Associated MuddyWater Threat Events



This figure illustrates the collection of threat events associated with MuddyWater, highlighting the relationship between multiple campaigns and shared indicators within MISP.

6.7 Malware, Tools, and Operational Tradecraft

The investigation identified MuddyWater as a threat actor that frequently relies on legitimate administrative utilities, scripting environments, and publicly available tools rather than exclusively using custom-developed malware. This approach reduces the likelihood of detection because many of the tools employed are commonly found within enterprise environments.

Commonly observed tools and malware associated with MuddyWater include:

Category	Examples
PowerShell	PowerShell-based downloaders, loaders, and execution frameworks
Remote Administration	Remote access tools (RATs) and PowerShell remoting
Living-off-the-Land Binaries (LOLBins)	powershell.exe, cmd.exe, certutil.exe, mshta.exe, rundll32.exe, regsvr32.exe
Persistence	Registry Run Keys, Scheduled Tasks, Startup folders

Category	Examples
Credential Access	Password theft and credential harvesting techniques
Command and Control	HTTP/HTTPS web protocols, encrypted communication channels

The extensive use of legitimate Windows components demonstrates MuddyWater's preference for "living off the land," allowing malicious activity to blend with normal administrative operations and complicating detection by traditional signature-based security controls.

6.8 Security Implications

The findings of this investigation highlight the importance of combining threat intelligence with continuous security monitoring. Because MuddyWater frequently adapts its infrastructure and operational techniques, organizations cannot rely solely on blocking known IOCs. Effective defense requires layered security controls, behavioral analytics, and intelligence-driven detection.

The intelligence collected during this investigation can be used to:

- Enrich SIEM alerts with contextual threat intelligence.
- Create custom detection rules for known MuddyWater behaviors.
- Improve threat hunting by searching for identified IOCs and TTPs.
- Strengthen incident response through faster attribution.
- Enhance network and endpoint monitoring with updated threat indicators.
- Support strategic risk assessments for organizations operating in targeted sectors.

7. MITRE ATT&CK Mapping and Threat Relationship Analysis

7.1 MITRE ATT&CK Framework Overview

The **MITRE ATT&CK (Adversarial Tactics, Techniques, and Common Knowledge)** framework is a globally recognized knowledge base that documents the real-world tactics, techniques, and procedures (TTPs) employed by cyber adversaries. Rather than focusing solely on malware or Indicators of Compromise (IOCs), the framework emphasizes adversary behavior throughout the attack lifecycle, enabling organizations to understand how attackers operate and to design effective detection, prevention, and response strategies.

Within the context of this investigation, MITRE ATT&CK provided a structured methodology for analyzing the operational behavior of the MuddyWater threat actor. By mapping observed activities to ATT&CK techniques, SecureOps analysts gained a deeper understanding of the

adversary's attack methodology and identified opportunities to improve detection engineering, threat hunting, and incident response capabilities.

The MISP Galaxy integration further enriched the investigation by linking MuddyWater intelligence directly to ATT&CK techniques, allowing analysts to correlate threat intelligence with standardized adversary behaviors.

7.2 Observed Tactics, Techniques, and Procedures (TTPs)

The intelligence collected during this investigation indicates that MuddyWater employs a combination of social engineering, legitimate administrative tools, malware, and stealth techniques to gain and maintain unauthorized access to victim environments.

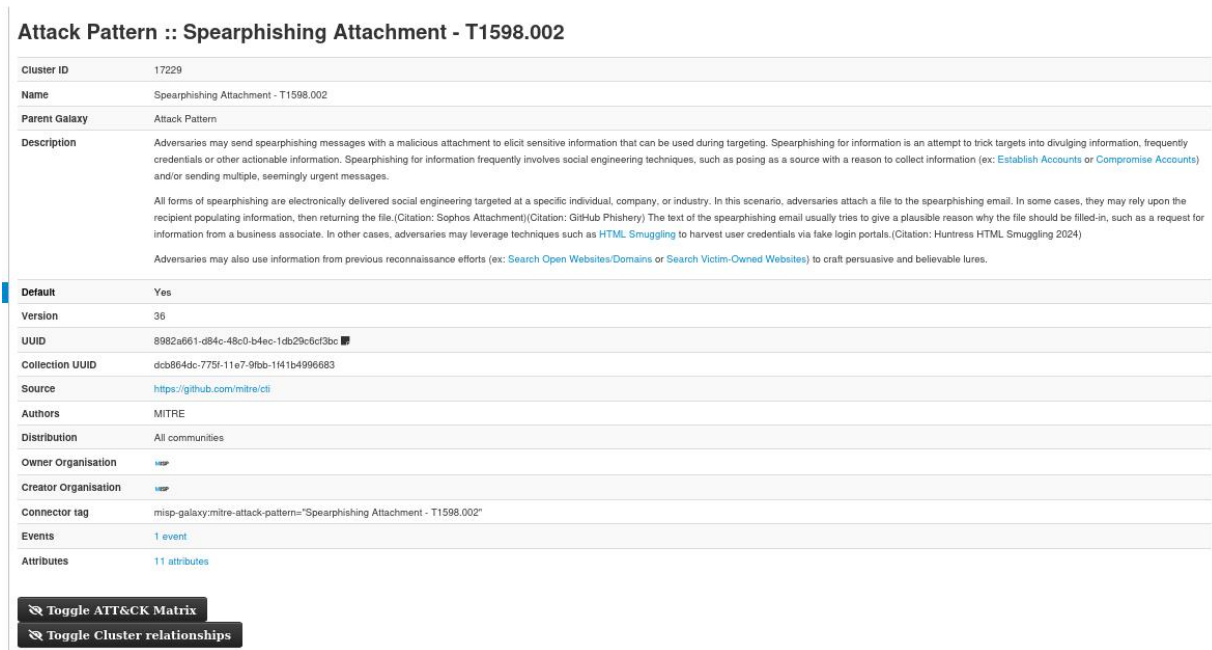
The table below summarizes the primary ATT&CK techniques associated with MuddyWater.

MITRE Tactic	Technique	Technique ID	Operational Purpose
Initial Access	Spear phishing Attachment	T1566.001	Deliver malicious attachments to targeted users.
Initial Access	Spear phishing Link	T1566.002	Redirect victims to credential harvesting or malware delivery sites.
Execution	PowerShell	T1059.001	Execute malicious scripts and download payloads.
Execution	Command and Scripting Interpreter	T1059	Execute attacker-controlled commands.
Persistence	Registry Run Keys / Startup Folder	T1547.001	Maintain persistence across system reboots.
Defense Evasion	Obfuscated Files or Information	T1027	Hide malicious payloads from detection.
Discovery	System Information Discovery	T1082	Gather information about compromised hosts.
Discovery	File and Directory Discovery	T1083	Enumerate sensitive files and directories.
Credential Access	OS Credential Dumping	T1003	Attempt to obtain stored credentials.

MITRE Tactic	Technique	Technique ID	Operational Purpose
Collection	Archive Collected Data	T1560	Compress and prepare stolen information for exfiltration.
Command and Control	Application Layer Protocol (Web Protocols)	T1071.001	Communicate with command-and-control infrastructure over HTTP/HTTPS.

These techniques illustrate MuddyWater's preference for blending malicious activity with legitimate administrative operations, making behavioral detection significantly more effective than relying solely on static signatures.

Figure 17. MITRE ATT&CK Mapping within MISP



This figure illustrates the MITRE ATT&CK mapping associated with the investigated threat intelligence. Mapping observed adversary behavior to standardized ATT&CK techniques enables analysts to better understand the attack lifecycle and develop more effective detection and response strategies.

7.3 Threat Relationship Analysis

One of the distinguishing features of MISP is its ability to correlate related threat intelligence objects and visualize relationships between threat actors, malware, campaigns, indicators, and attack techniques.

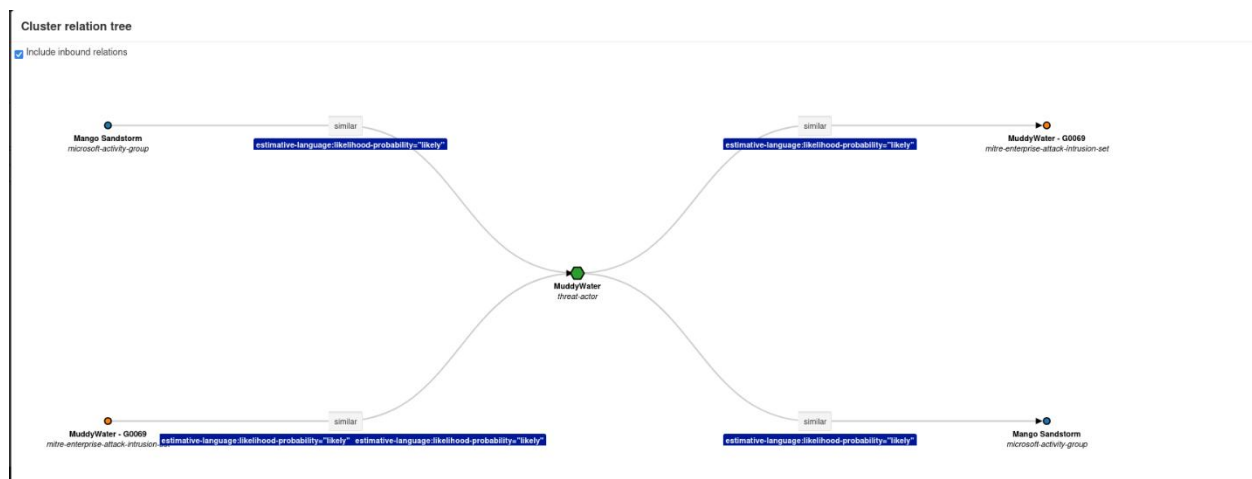
During the investigation, MISP's relationship graph revealed associations between MuddyWater and multiple intelligence entities, allowing analysts to move beyond isolated indicators and understand the broader operational ecosystem of the threat actor.

Relationship analysis identified connections among:

- Threat actors
- Related intelligence events
- Campaigns
- Indicators of Compromise (IOCs)
- ATT&CK techniques
- Infrastructure
- Malware
- Galaxy clusters

These correlations provide valuable context that supports attribution, prioritization, and strategic decision-making. For example, when the same infrastructure or malware family appears across multiple events, analysts can more confidently associate new activity with an existing adversary.

Figure 18. MuddyWater Threat Relationship Graph



This figure presents the relationship graph generated within MISP, illustrating how the MuddyWater threat actor is connected to related events, campaigns, indicators, and other

intelligence objects. Such visual correlations help analysts identify patterns that may not be apparent when reviewing individual indicators in isolation.

7.4 Operational Value for Security Operations Centers (SOC)

The intelligence gathered through this investigation provides practical value for Security Operations Centers by supporting several operational activities:

- **Threat Detection:** Incorporate extracted IOCs into SIEM platforms, IDS/IPS solutions, and endpoint security tools to improve detection of known malicious activity.
- **Threat Hunting:** Use observed TTPs and ATT&CK mappings to proactively search for behavioral evidence of MuddyWater activity, even when specific IOCs have changed.
- **Incident Response:** Accelerate investigation and containment by enriching alerts with contextual threat intelligence and known adversary behaviors.
- **Detection Engineering:** Develop behavioral detection rules that focus on suspicious PowerShell execution, persistence mechanisms, and command-and-control communications.
- **Threat Intelligence Sharing:** Share validated intelligence internally and with trusted partners to strengthen collective defense against evolving nation-state threats.

By integrating structured threat intelligence into daily SOC operations, organizations can move from a reactive security posture to a more proactive and intelligence-driven approach.

8. Key Findings

8.1 Investigation Summary

The primary objective of this project was to deploy and utilize the **Malware Information Sharing Platform (MISP)** to investigate and profile the **MuddyWater Advanced Persistent Threat (APT)** group through the collection, correlation, and analysis of structured cyber threat intelligence.

The investigation successfully demonstrated how a centralized Threat Intelligence Platform (TIP) can transform raw threat data into actionable intelligence that supports Security Operations Center (SOC) operations, threat hunting, incident response, and strategic cybersecurity decision-making.

Using synchronized community threat intelligence feeds, analysts successfully identified MuddyWater-related events, correlated Indicators of Compromise (IOCs), examined adversary relationships, and analyzed the threat actor's tactics, techniques, and procedures (TTPs). The

integration of MISP Galaxy clusters and the MITRE ATT&CK framework further enriched the investigation by providing contextual intelligence that extended beyond individual indicators.

The investigation confirmed that structured threat intelligence significantly improves an organization's ability to understand adversary behavior, attribute malicious activity, and implement proactive defensive measures against sophisticated nation-state actors.

8.2 Major Findings

The investigation produced several significant findings that are directly relevant to SecureOps Cybersecurity Inc. and its clients.

Finding 1 – Successful Deployment of a Threat Intelligence Platform

The project successfully deployed the Malware Information Sharing Platform (MISP) using a Docker-based containerized architecture on Kali Linux. The deployment demonstrated that MISP can be implemented efficiently within a secure and isolated environment while providing a scalable platform for intelligence collection, correlation, and sharing.

The successful deployment establishes a repeatable framework that can be adapted for future threat intelligence investigations involving other Advanced Persistent Threat (APT) groups.

Finding 2 – Effective Intelligence Collection through Community Feeds

The configured MISP community feeds successfully imported structured cyber threat intelligence containing threat events, malicious infrastructure, threat actor information, malware intelligence, and Indicators of Compromise.

By enabling trusted intelligence feeds, SecureOps analysts gained immediate access to continuously updated threat intelligence without relying solely on internally generated indicators.

This capability significantly improves intelligence freshness and expands visibility into emerging cyber threats.

Finding 3 – Comprehensive Threat Actor Attribution

The investigation successfully identified MuddyWater as the primary threat actor associated with the analyzed intelligence.

The collected intelligence confirmed multiple aliases used by leading cybersecurity organizations, including:

- Mango Sandstorm
- MERCURY

- Seedworm
- Static Kitten
- TEMP.Zagros
- Boggy Serpens
- TA450

Understanding these naming conventions improves cross-vendor intelligence correlation and reduces ambiguity during investigations.

Finding 4 – Identification of High-Value Indicators of Compromise

The investigation identified multiple categories of Indicators of Compromise associated with MuddyWater operations, including:

- Malicious IP addresses
- Domains
- URLs
- Registry artifacts
- File hashes
- Malware-related indicators
- Threat actor associations

These indicators can be integrated into organizational security technologies to improve threat detection, accelerate investigations, and strengthen defensive monitoring.

Finding 5 – Correlation of Related Threat Intelligence Events

MISP's event correlation capabilities enabled analysts to identify relationships between multiple threat intelligence events linked to MuddyWater.

Rather than examining isolated indicators, analysts were able to observe recurring operational patterns, infrastructure reuse, and campaign relationships that provided greater confidence in threat attribution.

Finding 6 – Mapping Adversary Behavior to MITRE ATT&CK

The investigation successfully mapped observed adversary behaviors to the MITRE ATT&CK framework.

This mapping provides defenders with a standardized understanding of MuddyWater's operational techniques and supports:

- Threat hunting
- Detection engineering
- Security monitoring
- Incident response
- Defensive planning

Behavior-based detection remains one of the most effective strategies for identifying advanced threat actors that frequently modify their infrastructure.

Finding 7 – Visualization of Threat Relationships

The use of MISP Galaxy clusters and relationship graphs enabled analysts to visualize connections between:

- Threat actors
- Intelligence events
- Indicators
- Campaigns
- ATT&CK techniques
- Supporting intelligence objects

This contextual visualization greatly enhanced situational awareness and simplified the analysis of complex intelligence relationships.

8.3 Business Impact

From a business perspective, the investigation demonstrates how structured threat intelligence can directly improve SecureOps Cybersecurity Inc.'s operational capabilities.

Key organizational benefits include:

- Faster identification of emerging threats.
- Improved threat attribution through intelligence correlation.

- Enhanced SOC investigation workflows.
- Reduced analyst investigation time.
- Improved quality of intelligence reports delivered to clients.
- Better protection of critical infrastructure customers.
- Increased customer confidence through intelligence-driven security services.

The project also reinforces SecureOps' strategic objective of providing proactive cybersecurity services rather than relying solely on reactive incident response.

8.4 Operational Value

The intelligence generated throughout this investigation can be operationalized across multiple cybersecurity functions.

Security Operations Center (SOC)

- IOC enrichment
- Alert validation
- Threat prioritization
- Incident triage

Threat Hunting

- Search for MuddyWater TTPs
- Behavioral analytics
- Historical IOC searches
- Infrastructure correlation

Detection Engineering

- SIEM detection rules
- EDR detections
- IDS/IPS signatures
- Firewall blocklists

Incident Response

- Rapid threat attribution
- IOC validation
- Malware investigation
- Containment support

Threat Intelligence

- Intelligence sharing
- Campaign tracking
- Threat actor monitoring
- Future intelligence enrichment

8.5 Lessons Learned

The investigation produced several important technical and strategic lessons.

Threat intelligence provides context.

Raw Indicators of Compromise become significantly more valuable when enriched with information regarding threat actors, campaigns, attack techniques, and infrastructure relationships.

Correlation improves attribution.

MISP's ability to correlate indicators across multiple intelligence sources enables analysts to develop higher-confidence threat assessments than would be possible using isolated indicators.

Community intelligence accelerates investigations.

Trusted threat intelligence feeds provide timely information regarding emerging threats and reduce the time required to identify known adversaries.

ATT&CK mapping strengthens defensive capabilities.

Mapping adversary behaviors to the MITRE ATT&CK framework enables defenders to move beyond signature-based detection and implement behavior-based monitoring strategies.

Intelligence supports proactive defense.

Threat intelligence should be viewed as a continuous operational capability rather than a one-time investigative activity. Continuous intelligence collection and analysis improve an organization's ability to anticipate, detect, and respond to evolving cyber threats.

Section Summary

This investigation successfully demonstrated the practical application of MISP as a Threat Intelligence Platform for collecting, correlating, and analyzing cyber threat intelligence associated with the MuddyWater Advanced Persistent Threat group. The findings confirm that integrating structured threat intelligence into SOC operations enhances visibility, improves attribution, and strengthens an organization's ability to defend against sophisticated nation-state adversaries.

9. Security Recommendations

9.1 Overview

The findings of this investigation demonstrate that the evolving tactics employed by nation-state Advanced Persistent Threat (APT) groups such as MuddyWater require organizations to adopt an intelligence-driven cybersecurity strategy. Traditional perimeter-based security controls are no longer sufficient to defend against adversaries that employ legitimate administrative tools, sophisticated phishing campaigns, credential theft, and long-term persistence.

Based on the intelligence gathered throughout this investigation, the following recommendations are proposed to strengthen SecureOps Cybersecurity Inc.'s ability to detect, investigate, and respond to future threats while enhancing the overall security posture of its clients.

9.2 Strategic Recommendations

Recommendation 1 – Expand Threat Intelligence Integration

Threat intelligence should become a continuous operational capability rather than an activity performed only after security incidents.

SecureOps should continue integrating multiple trusted intelligence sources into MISP to improve visibility into emerging cyber threats and enrich existing intelligence with additional context.

Recommended actions include:

- Integrate commercial threat intelligence feeds alongside community feeds.
- Regularly synchronize threat intelligence to maintain current IOC datasets.
- Participate in trusted intelligence-sharing communities.
- Develop internal intelligence repositories for sector-specific threats.

Expected benefits include improved threat attribution, broader intelligence coverage, and faster identification of emerging attack campaigns.

Recommendation 2 – Establish Continuous Threat Actor Monitoring

Threat actors continually evolve their infrastructure, malware, and operational techniques.

SecureOps should establish continuous monitoring of high-priority threat actors such as:

- MuddyWater
- APT29
- APT28
- Lazarus Group
- Sandworm
- Volt Typhoon

Continuous monitoring enables analysts to identify changes in adversary behavior before attacks affect client environments.

9.3 Operational Recommendations

Recommendation 3 – Integrate MISP with Security Operations Center (SOC) Platforms

Threat intelligence becomes significantly more valuable when integrated directly into operational security workflows.

SecureOps should integrate MISP with technologies such as:

- SIEM platforms
- Endpoint Detection and Response (EDR)
- Security Orchestration, Automation, and Response (SOAR)
- Intrusion Detection and Prevention Systems (IDS/IPS)
- Firewall Management Platforms

Automated IOC enrichment allows analysts to receive contextual intelligence during investigations rather than manually researching indicators.

Recommendation 4 – Develop IOC-Based Detection Rules

Indicators of Compromise identified during this investigation should be operationalized within enterprise security solutions.

Detection content should include:

- Malicious IP addresses
- Malicious domains
- URLs
- File hashes
- Registry artifacts
- Email indicators
- Command-and-control infrastructure

Security teams should continuously validate these indicators because threat actors frequently rotate infrastructure.

Recommendation 5 – Adopt Behavioral Detection

Since MuddyWater frequently abuses legitimate Windows administrative tools, organizations should prioritize behavioral detection over signature-based detection.

Detection strategies should focus on:

- Unusual PowerShell execution
- Encoded PowerShell commands
- Suspicious use of LOLBins
- Registry persistence modifications
- Scheduled task creation
- Unusual outbound HTTPS communications
- Credential dumping behavior

Behavioral analytics significantly improve detection against advanced adversaries that continuously modify malware signatures.

9.4 Threat Hunting Recommendations

Threat hunting activities should be performed regularly using intelligence gathered from MISP.

Recommended hunting scenarios include:

- Search for PowerShell activity associated with MuddyWater tradecraft.
- Hunt for suspicious parent-child process relationships.
- Monitor abnormal network connections to external infrastructure.
- Identify persistence mechanisms established through registry modifications or scheduled tasks.
- Investigate unusual authentication activity indicating credential theft.

Threat hunting should combine IOC-based searches with ATT&CK-based behavioral analytics to improve detection effectiveness.

9.5 User Awareness Recommendations

Human error remains one of the primary attack vectors exploited by Advanced Persistent Threat groups.

SecureOps should recommend that client organizations:

- Conduct regular phishing awareness training.
- Implement simulated phishing exercises.
- Train employees to identify malicious attachments and links.
- Encourage timely reporting of suspicious emails.
- Promote secure password management and multi-factor authentication.

Improving user awareness reduces the likelihood of successful initial access through social engineering attacks.

9.6 Infrastructure Security Recommendations

To reduce the attack surface exploited by MuddyWater and similar threat actors, organizations should strengthen infrastructure security by implementing the following controls:

- Enable Multi-Factor Authentication (MFA) for all privileged and remote access accounts.
- Enforce the principle of least privilege.
- Segment critical networks to limit lateral movement.
- Maintain a robust vulnerability management and patching program.
- Disable unnecessary administrative services.
- Restrict PowerShell execution through appropriate execution policies and application control.
- Monitor outbound network traffic for anomalous command-and-control communications.
- Maintain secure offline backups and regularly test restoration procedures.

9.7 MITRE ATT&CK–Aligned Defensive Recommendations

The ATT&CK mappings identified during this investigation can be translated directly into defensive controls.

Observed ATT&CK Technique	Recommended Defensive Measure
Spearphishing	Advanced email security, attachment sandboxing, phishing awareness training
PowerShell Execution	PowerShell logging, Constrained Language Mode, AMSI integration
Registry Persistence	Continuous registry monitoring and change detection
Credential Access	Credential Guard, MFA, privileged access management
Command and Control	DNS filtering, proxy inspection, outbound traffic monitoring
Discovery Techniques	Endpoint monitoring, EDR telemetry, anomaly detection
Living-off-the-Land Techniques	Application allowlisting and behavioral detection for native system tools

Aligning security controls with ATT&CK techniques enables organizations to move from reactive detection to a proactive defense strategy focused on adversary behavior.

9.8 Recommendations for Future Enhancements

While this project successfully demonstrated the use of MISP for threat intelligence analysis, several enhancements could further strengthen SecureOps' capabilities:

- Integrate MISP with **Wazuh**, **Splunk**, or **Microsoft Sentinel** for automated IOC enrichment and alert correlation.
- Use the **MISP REST API** and **PyMISP** to automate threat intelligence collection, analysis, and reporting.
- Incorporate **YARA** rules for malware identification and classification.
- Expand intelligence sources by integrating commercial and sector-specific threat feeds.
- Automate IOC validation using services such as **VirusTotal**, **AbuseIPDB**, and **URLhaus**.
- Develop threat hunting playbooks based on observed MuddyWater tactics.
- Build executive dashboards that visualize threat actor trends, ATT&CK coverage, and IOC activity for security leadership.

Section Summary

The recommendations presented in this report are intended to enhance SecureOps Cybersecurity Inc.'s threat intelligence maturity and improve its ability to defend clients against sophisticated nation-state adversaries. By combining structured threat intelligence, behavioral analytics, MITRE ATT&CK mapping, and continuous monitoring, the organization can reduce detection time, improve incident response effectiveness, and strengthen resilience against evolving cyber threats. Implementing these recommendations will also support a more proactive, intelligence-driven security posture capable of adapting to the rapidly changing threat landscape.

10. Conclusion

10.1 Project Conclusion

The **MuddyWater Threat Intelligence Investigation Using the Malware Information Sharing Platform (MISP)** successfully demonstrated how structured cyber threat intelligence can be leveraged to identify, profile, and analyze sophisticated nation-state adversaries. By deploying MISP within a Docker-based environment and integrating trusted community threat

intelligence feeds, SecureOps Cybersecurity Inc. established a centralized platform for collecting, correlating, and enriching cyber threat intelligence.

The investigation addressed the organization's business challenge of improving visibility into increasingly sophisticated attacks targeting government agencies and critical infrastructure. Through the analysis of synchronized threat intelligence feeds, the project successfully identified the MuddyWater Advanced Persistent Threat (APT) group, developed a comprehensive threat actor profile, examined associated campaigns, extracted Indicators of Compromise (IOCs), and mapped observed adversary behaviors to the MITRE ATT&CK framework.

Rather than relying solely on isolated indicators, the project demonstrated the value of contextual intelligence. MISP's correlation engine and Galaxy clusters enabled analysts to establish relationships between threat actors, infrastructure, campaigns, malware, and attack techniques, resulting in higher-confidence threat attribution and more meaningful intelligence products. This contextual understanding significantly enhances the ability of Security Operations Center (SOC) teams to prioritize alerts, conduct threat hunting, and accelerate incident response.

The investigation also highlighted the operational value of integrating cyber threat intelligence into everyday security operations. The extracted IOCs, ATT&CK mappings, and threat actor profiles provide reusable intelligence that can be incorporated into SIEM platforms, Endpoint Detection and Response (EDR) solutions, Intrusion Detection Systems (IDS), and Security Orchestration, Automation, and Response (SOAR) workflows. This enables organizations to move beyond reactive incident handling toward a proactive, intelligence-driven security model capable of anticipating and mitigating emerging threats.

Furthermore, the project reinforced the importance of collaborative intelligence sharing within the cybersecurity community. By leveraging trusted MISP feeds, SecureOps benefited from continuously updated threat intelligence contributed by multiple organizations, improving the quality, timeliness, and accuracy of its investigations. This collaborative approach is essential for defending against modern threat actors that frequently change infrastructure, malware, and operational techniques to evade detection.

From a professional development perspective, this project provided practical experience in deploying and managing a Threat Intelligence Platform, configuring Docker-based cybersecurity infrastructure, analyzing real-world threat intelligence, profiling Advanced Persistent Threat groups, and applying the MITRE ATT&CK framework to understand adversary behavior. These are core competencies expected of Threat Intelligence Analysts, SOC Analysts, Incident Responders, and Cyber Threat Hunters working within modern cybersecurity operations.

Overall, the project successfully achieved its stated objectives and demonstrated that intelligence-driven cybersecurity significantly improves an organization's ability to understand adversaries, strengthen defensive capabilities, and support informed decision-making. The knowledge, methodologies, and investigative processes developed throughout this project

provide a solid foundation for future threat intelligence investigations involving other Advanced Persistent Threat groups and emerging cyber threats.

10.2 Project Outcomes

The investigation achieved the following outcomes:

- Successfully deployed the Malware Information Sharing Platform (MISP) using Docker.
- Configured and synchronized trusted community threat intelligence feeds.
- Investigated and profiled the MuddyWater Advanced Persistent Threat (APT) group.
- Identified and analyzed Indicators of Compromise (IOCs) associated with MuddyWater campaigns.
- Examined related threat events and intelligence relationships using MISP correlation features.
- Mapped observed adversary behaviors to the MITRE ATT&CK framework.
- Produced actionable threat intelligence to support Security Operations Center (SOC) operations.
- Developed strategic recommendations to improve organizational cyber resilience.
- Created a repeatable methodology for future threat intelligence investigations.

10.3 Future Work

While the objectives of this project were successfully achieved, there are several opportunities for future enhancement that would further increase the operational value of the solution.

Future developments may include:

- Integrating MISP with SIEM platforms such as **Splunk**, **Microsoft Sentinel**, or **Wazuh** to automate IOC enrichment and alert correlation.
- Leveraging the **MISP REST API** and **PyMISP** to automate threat intelligence collection, indicator extraction, and reporting.
- Expanding intelligence sources by incorporating commercial threat intelligence feeds and industry-specific information-sharing communities.
- Implementing automated malware analysis and sandbox integrations to enrich threat intelligence with behavioral insights.

- Developing executive dashboards that visualize threat actor activity, ATT&CK technique coverage, IOC trends, and campaign timelines.
- Extending the investigation methodology to additional Advanced Persistent Threat groups, ransomware operators, and financially motivated cybercriminal organizations.

These enhancements would further improve SecureOps' ability to deliver timely, intelligence-driven cybersecurity services while strengthening the resilience of its clients against evolving cyber threats.

10.4 Final Statement

Cybersecurity is no longer solely about responding to attacks after they occur; it is about understanding adversaries before they strike. Threat intelligence transforms isolated technical artifacts into meaningful knowledge that enables organizations to anticipate, detect, and mitigate cyber threats with greater speed and confidence.

This project demonstrates that the Malware Information Sharing Platform (MISP), when combined with structured threat intelligence, collaborative information sharing, and frameworks such as MITRE ATT&CK, provides a powerful capability for identifying sophisticated threat actors, understanding their operational behavior, and supporting informed defensive decision-making. As cyber threats continue to evolve, intelligence-driven security operations will remain a critical component of protecting organizations, critical infrastructure, and national interests against increasingly sophisticated adversaries.

11. References

The following references were used to support the deployment, investigation, threat intelligence analysis, and threat actor profiling conducted throughout this project. They include authoritative sources from cybersecurity organizations, government agencies, and industry-recognized threat intelligence providers.

Academic and Technical References (APA 7th Edition)

CISA. (2024). *Iranian state-sponsored actors*. Cybersecurity and Infrastructure Security Agency. <https://www.cisa.gov>

CISA, FBI, NSA, & U.S. Cyber Command. (2023). *Iranian state-sponsored cyber actors conduct cyber operations against critical infrastructure*. <https://www.cisa.gov/news-events/cybersecurity-advisories>

Docker Inc. (2025). *Docker documentation*. <https://docs.docker.com>

FIRST. (2024). *Traffic Light Protocol (TLP)*. <https://www.first.org/tlp>

Git. (2025). *Git documentation*. <https://git-scm.com/docs>

MITRE Corporation. (2025). *MITRE ATT&CK® knowledge base*. <https://attack.mitre.org>

MITRE Corporation. (2025). *ATT&CK Enterprise Matrix*. <https://attack.mitre.org/matrices/enterprise>

Microsoft Threat Intelligence. (2024). *Mango Sandstorm*. <https://www.microsoft.com/security/blog>

MISP Project. (2025). *Malware Information Sharing Platform & Threat Sharing documentation*. <https://www.misp-project.org>

MISP Project. (2025). *MISP galaxies documentation*. <https://www.misp-project.org/galaxy.html>

National Institute of Standards and Technology. (2024). *NIST Cybersecurity Framework (CSF) 2.0*. <https://www.nist.gov/cyberframework>

Open Container Initiative. (2024). *Container standards*. <https://opencontainers.org>

The MITRE Corporation. (2025). *Common Attack Pattern Enumeration and Classification (CAPEC)*. <https://capec.mitre.org>

Threat Intelligence Sources

The following threat intelligence sources were utilized during the investigation through the Malware Information Sharing Platform (MISP).

- CIRCL OSINT Feed
- Botvrij Threat Feed
- ELLIO Threat Feed

These feeds supplied the Indicators of Compromise (IOCs), threat actor intelligence, campaign information, and event correlation data analyzed throughout this project.

Threat Intelligence Frameworks

The following frameworks and intelligence models informed the analysis and interpretation of threat actor behavior:

- MITRE ATT&CK® Framework
- MISP Galaxy Threat Actor Taxonomy
- Cyber Kill Chain (conceptual reference)
- Diamond Model of Intrusion Analysis (conceptual reference)
- NIST Cybersecurity Framework (CSF) 2.0

Software and Technologies

The following technologies were implemented during the project:

Technology	Purpose
Kali Linux	Investigation workstation
Docker	Containerization platform
Docker Compose	MISP deployment orchestration
Git	Version control and repository management
MISP	Threat Intelligence Platform
MISP Galaxy	Threat actor enrichment and relationship analysis

Technology Purpose

MITRE ATT&CK Adversary behavior mapping

Note: Although the original project proposal listed **Python** and **Elasticsearch** in the intended technology stack, the implementation documented in this report focused on deploying and using MISP through Docker on Kali Linux. Since no Python automation or Elasticsearch integration was implemented as part of the investigation, they have not been listed as deployed technologies in this report. This keeps the documentation aligned with the actual work performed.

Images and Screenshots

All screenshots included in this report were captured by the author during the implementation and investigation phases of this project using the Malware Information Sharing Platform (MISP).

The screenshots document:

- MISP deployment
- Docker implementation
- Threat feed configuration
- Threat intelligence synchronization
- MuddyWater event investigation
- Threat actor profiling
- Indicators of Compromise (IOCs)
- Threat relationship analysis
- MITRE ATT&CK mappings

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